

**KISI-KISI SOAL / DESKRIPSI TEKNIS
LOMBA KOMPETENSI SISWA (LKS) SMK
TINGKAT PROVINSI JAWA TENGAH
TAHUN 2024**

BIDANG LOMBA :

Cyber Security

Soal dan tantangan yang diujikan pada kompetisi LKS bidang lomba Cyber Security bersifat rahasia, namun sesuai dengan kategori/aspek/kelemahan yang tertera pada silabus ini. Kompetitor dapat fokus mempelajari kategori/aspek/kelemahan yang tertera dibawah ini.

Web Exploitation

- Form Injection (e.g. File Upload)
- Session Injection & Broken Access Control
- Business Logic Error
- SQLi
- Blind SQLi
- LFI
- RFI
- SSTI
- XSS
- SSRF
- Object Deserialization
- RCE

Binary Exploitation

- Buffer overflow
- Integer overflow / underflow
- Shellcode
- Format String
- ROP chain (ret2libc, ret2win, dll)
- Stack Pivoting
- Bypass protection (PIE, CANARY, NX, Relro)

Reverse Engineering

- Run Program (**ELF/EXE**)
- Strings, Pipe (`|`), Grep
- Static Analysis (**Reconstruct** Algorithm)
- Dynamic Analysis (**Tracing, GDB**)
- Low Level File Formats (**Assembly & Bytecodes** Translation)
- Compiled Programming Language Syntax Format in Executable (**C,C++**)
- Arsitektur : **x86, x86_64** (32/64 bit)
- **Obfuscation (Known/Custom Encryption) & Simple Binary Patching**

Forensic

- Steganography
- Exiftool & Strings (Metadata)
- File Carving (binwalk, foremost, photorec)
- Network Forensic (PCAP/PCAPNG)
- OS Forensic (Browser Forensic, AppData Forensic, Third Party App Forensic, Digital Artifact Discovery <- Only Windows OS)
- Malware Analysis

Cryptography

- Classical ciphers (contoh: Vigenere, Caesar, Atbash, Affine, Substitution, XOR)
- Attack on RSA (contoh: Hastad, common modulus attack, twin prime, multiprimes)
- Attack on AES (contoh: serangan pada mode-mode ECB, CBC, OFB, CFB)
- Attack on ECC (contoh: Smart's attack)
- Attack on DSA (contoh: attack on ECDSA, attack on RSA signature)
- Hashing (contoh: length extension attack)

System Security (terkait Infrastructure Hardening dan Attack & Defense)

- VPN connection
- SSH connection
- CVE exploit and mitigation
- Linux-based OS administration (user, group, permissions, root access, package installation, etc.)
- Event and process monitoring
- Enumeration (port scanning, etc.)
- Data exfiltration
- Privilege escalation
- Firewall policy
- User account policy
- Source code review (PHP, Python, Java, C)
- Source code patching